

Annotation

Vladislav Goncharenko

Senior researcher



MIPT, spring 2026

Outline

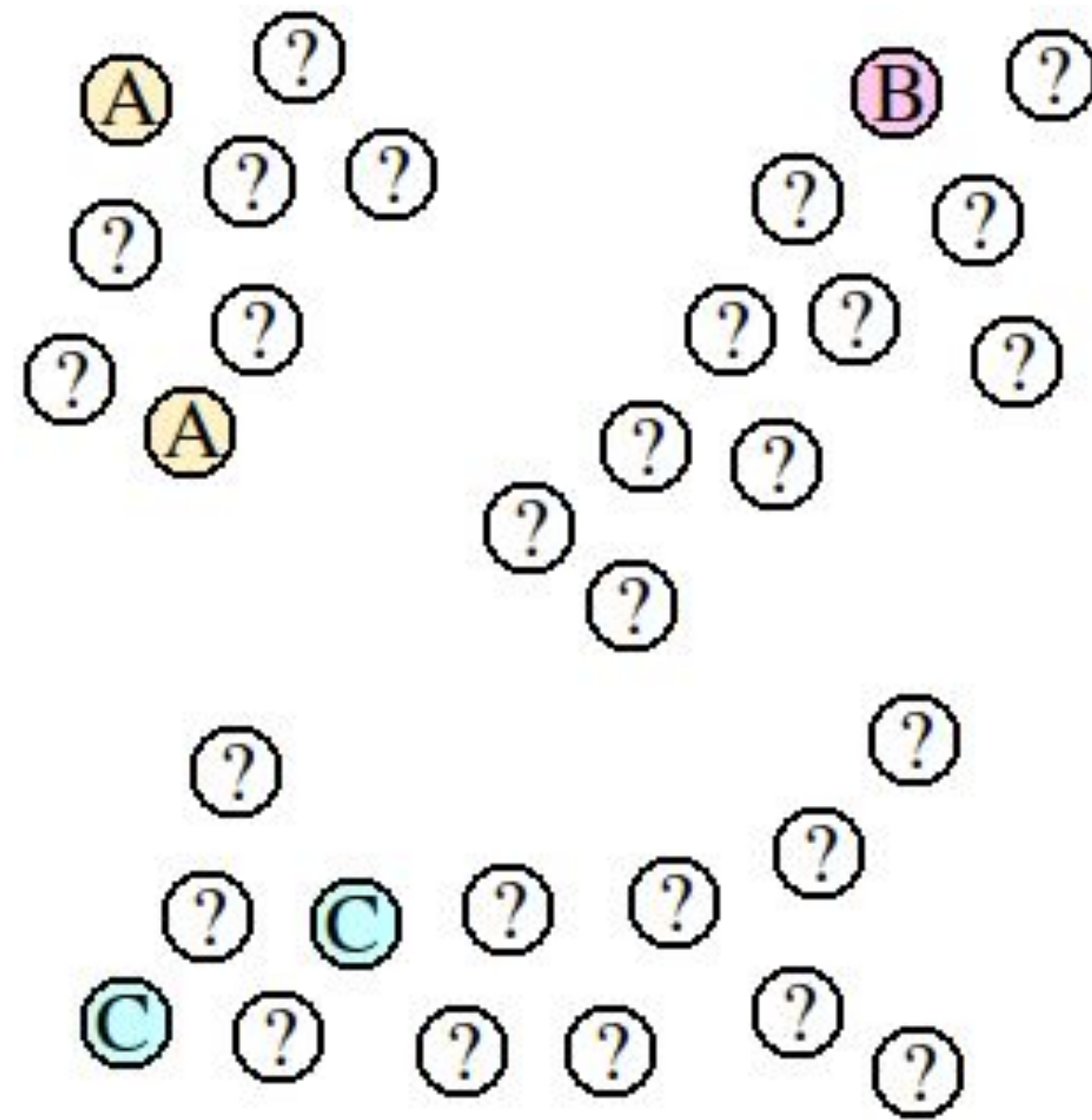
- Annotation pipeline
 - overlap
 - honeypots
- People management
 - inhouse
 - crowd
- Instructions
- Solutions for fields
 - CV
 - NLP

Types of problems in ML

- Supervised
 - Classification, Regression, Ranking
 - Segmentation, Object Detection, Speech2Text
- Unsupervised
 - Clustering, Dimensionality Reduction, Generation
- Self-supervised
 - Masked Language Modeling, Next Sentence Prediction, Image to Text
- many others
 - Semi-supervised, Reinforcement, Federated, ...

Transduction in ML

[https://en.wikipedia.org/wiki/Transduction_\(machine_learning\)](https://en.wikipedia.org/wiki/Transduction_(machine_learning))



What to annotate?

- Tabular data
 - classification, regression
- Texts
 - word type
- Images
 - segmentation, detection, skeletons, ocr, identification
- Videos
 - object tracks, crowd counting, video quality
- 3D data (point clouds, meshes)
 - 3d detection, segmentations, tracking
- Sounds
 - speech2text: subtitles, genre of music

What to annotate

- Tables
- Timeseries
- Text
- Images
 - classification, OCR
- Video
- 3d data
 - point clouds
 - meshes
- Sounds
 - Speech2text
 - segment classification
- Multimodal data



Task management

1. Annotation is not cheap
- 2.



Typical task for annotation

Realtime model assessment



Example: Relevance levels for search engine

query + document + metainfo

- vital
- rel+
- rel
- rel-
- not rel

[Google's guide for query annotation](#)



Task management



1. Owner
 - a. responsible for the whole pipeline
2. Experts
 - a. data engineers
 - b. ML engineers and analytics
 - c. model users (non technicians)
3. Leaders
 - a. Supervise assessors
 - b. label data
4. Assessors
 - a. label data by instruction

Who are the assessors?

Usually low paid people with low motivation and poor education.

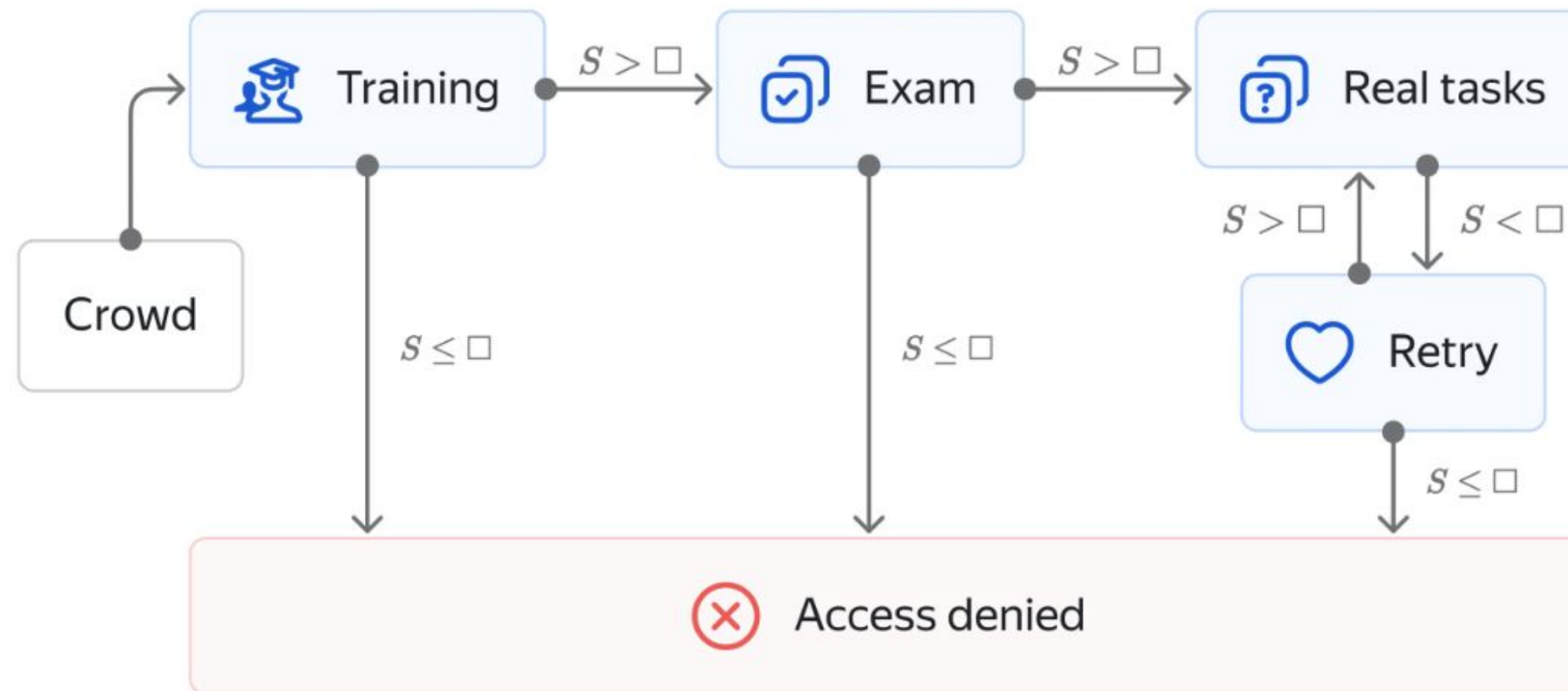
Usually retired or students

So they are not giving the best answers

Quality of labeling is unstable even in one person.



Assessor lifecycle



1. Learning
2. Qualification
3. Tasks completions
4. Requalification

To rank assessor we use skill level measuring our trust

Requalification can happen because of absence

Honeypots aka golden set

In computer terminology, a honeypot is a computer security mechanism set to detect, deflect, or, in some manner, counteract attempts at unauthorized use of information systems.

[Common knowledge site](#)

Honeypot rules

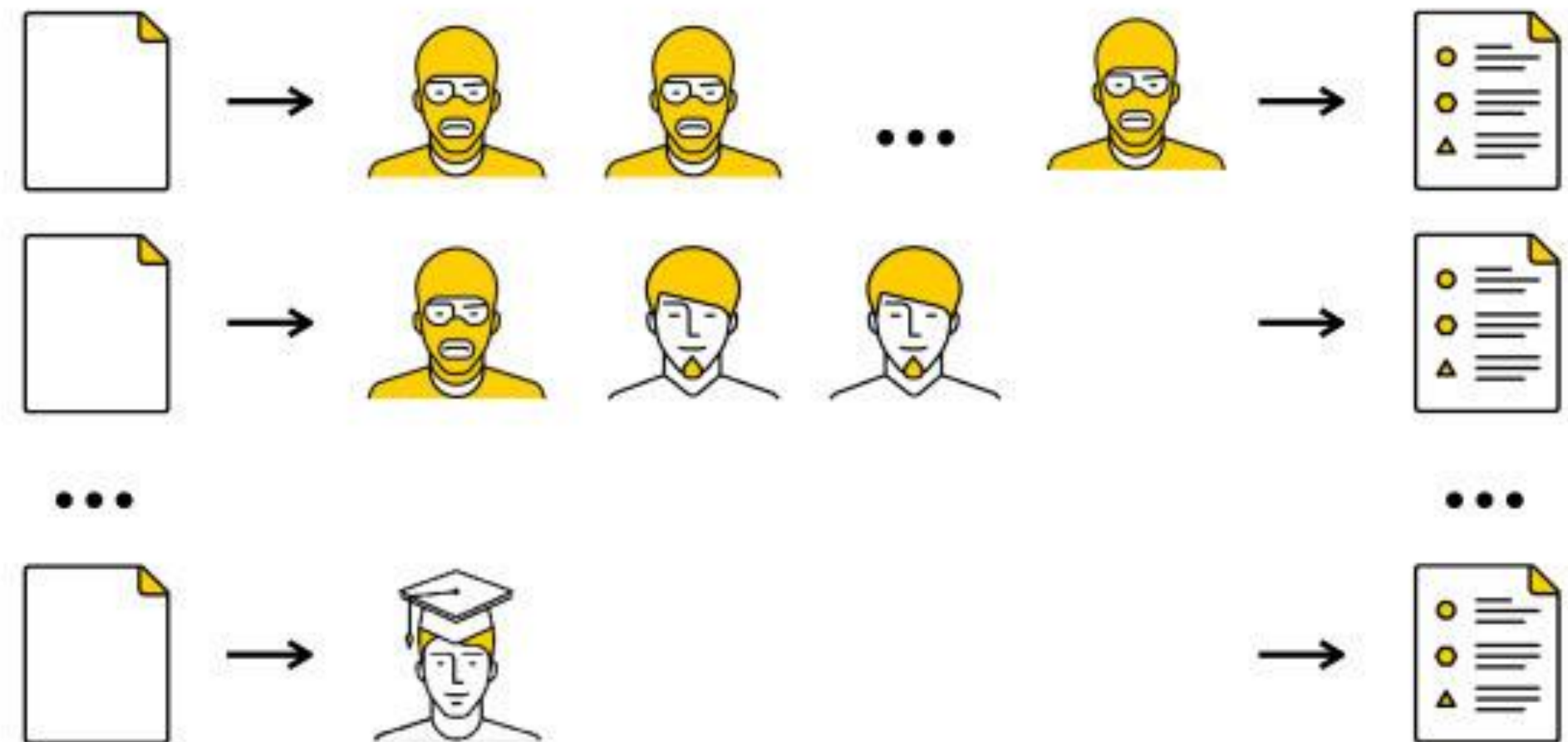
- regular rotation!!!
- building from lead assessors (more trusted)

Overlap

For best result we need to annotate the same sample several times.

Usually it's 3 people per sample with Majority Vote aggregation.

Also there are advanced aggregation techniques which requires setup for each single case

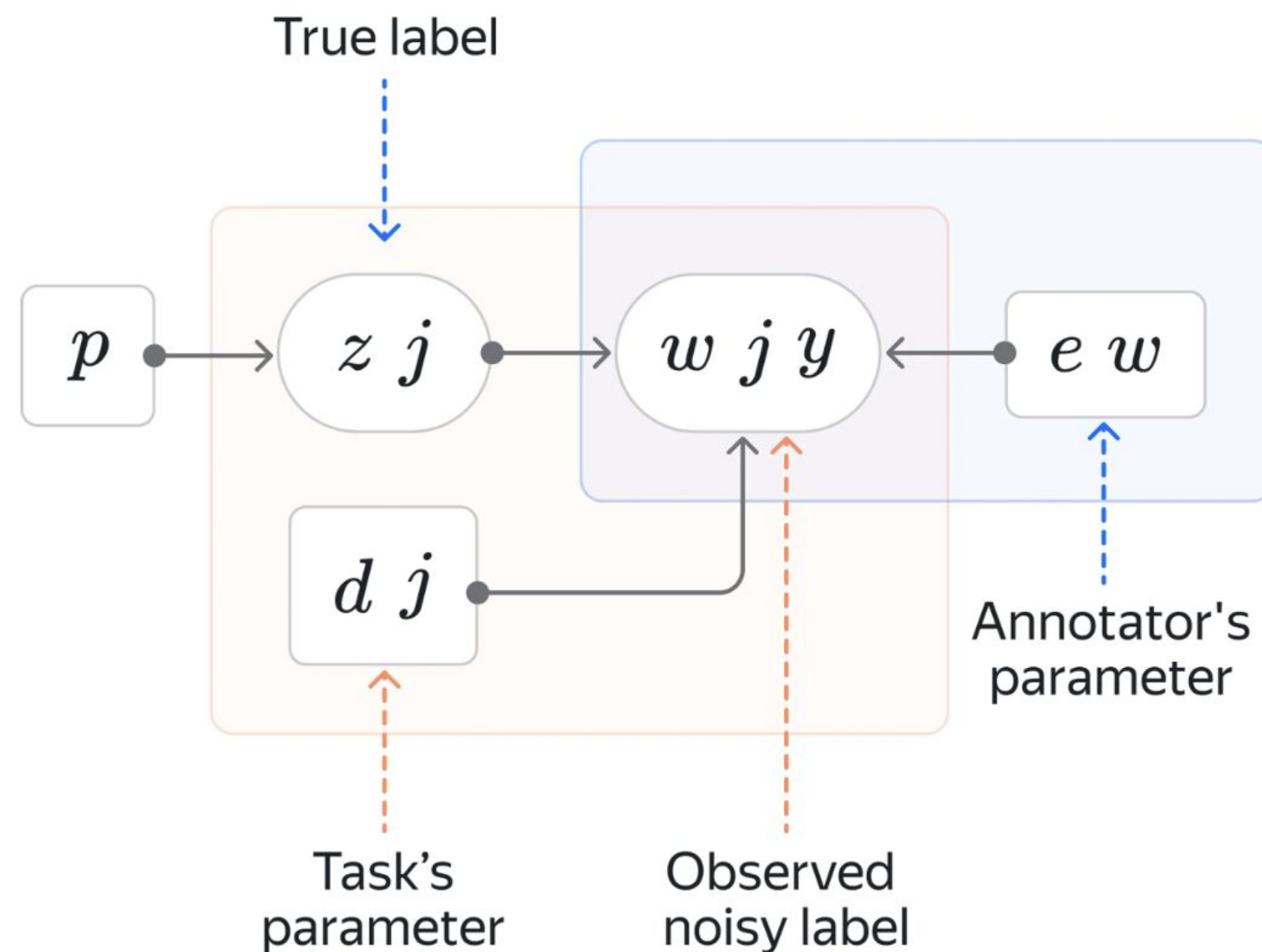


Aggregation

For best result we need to annotate the same sample several times.

Usually it's 3 people per sample aggregation.

Also there are advanced aggregation techniques which requires setup for each single case



Bidding

Quality of labeling barely depend on money you pay.

Higher prices will make it faster mostly.

How much to pay?

1. Budget we have and required amount of labels
2. Look for existing tasks
3. Ask LLM in Deep Research mode
4. Estimate it yourself based on average salary and time consumption
 - a. [Estimate learning duration](#)
 - b. Measure as experienced assessor
 - c. first as honest person
 - d. then as frauder

Project management

 **Project**

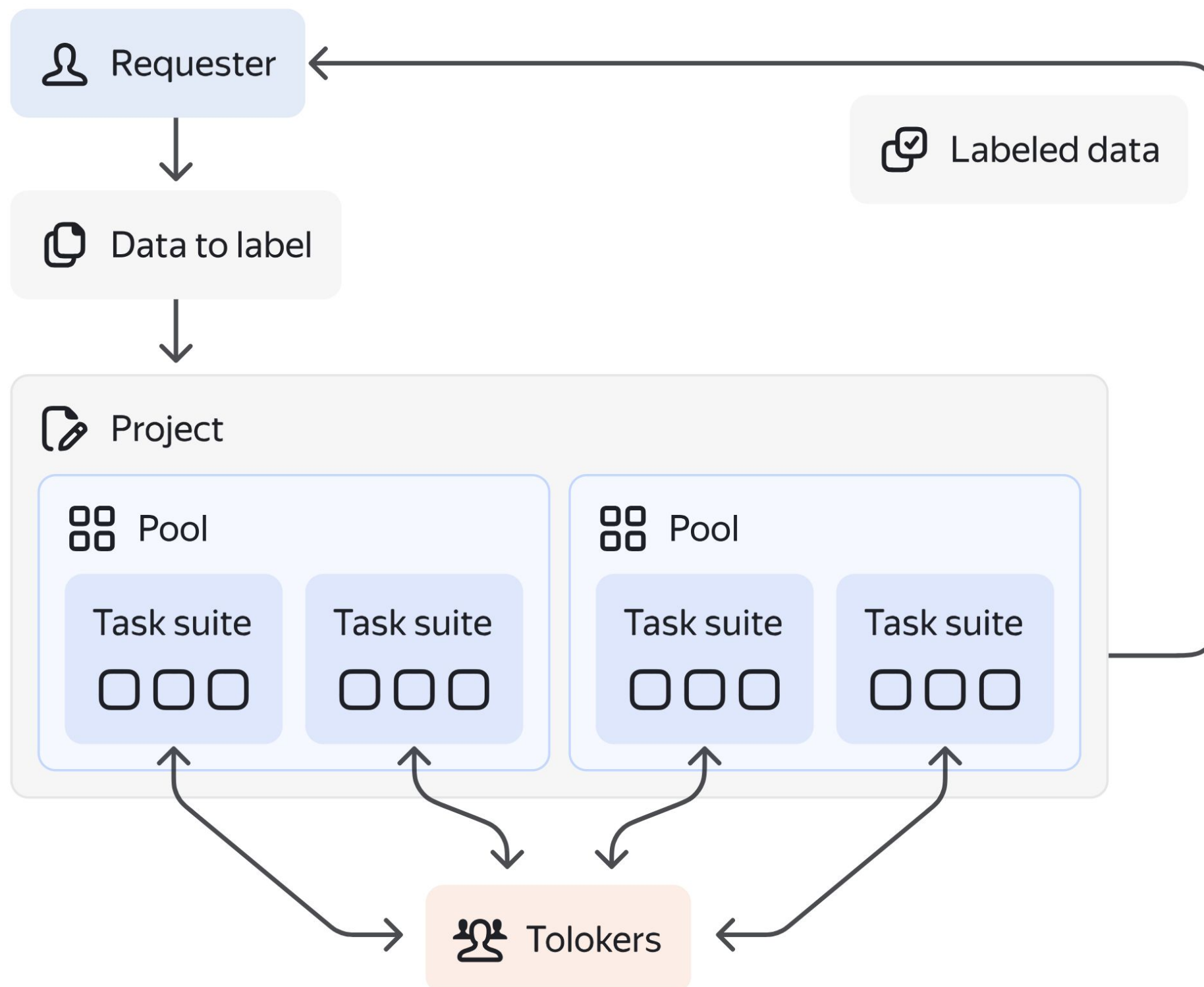
Main description
Interface
Task instructions
Input and output data specifications

 **Pool**

Price
Toloker selection
Overlap
Quality control

 **Task**

Task data
text, image URLs, and more
Correct responses
Tips for training



Platform types

	Inhouse	Outsource	Crowd
Scalability	low	medium	high
Quality	highest	high	low
Cost	moderate	high	low
Control	full	high	medium
Assessors	1-10	~100	>1000

- For crowd per task cost does not really matter after some threshold
- Experts should be employed full time, Assessors paid per label
- Don't use platform as database! Store your data in trusted sources

Crowd tools

They are proprietary

- Scale AI <https://scale.com/>
 - <https://scale.com/guides/data-labeling-annotation-guide>
- <https://www.appen.com/>
- <https://www.telusdigital.com/solutions/data-for-ai-training>
- Toloka <https://toloka.ai/>
 - <https://tasks.yandex.ru/>
- MTurk Amazon <https://www.mturk.com/>
- <https://roboflow.com/>
- crowds of them...

FOSS tools

Self hosted

- standalone
 - [VIA](#)
- server based
 - [CVAT](#)
 - <https://github.com/HumanSignal/label-studio>
 - <https://github.com/wkentaro/labelme>
- lots of them

Overviews: [one](#), [two](#)

Toloka

<https://www.youtube.com/watch?v=gkjOH5SYkuU&list=PLJOzdkh8T5kq9eSxQhUJ63HH68oPj532O>

tasks available in toloka <https://toloka.ai/docs/tutorials/usecases/>

structure <https://toloka.ai/docs/guide/overview/>
<https://toloka.ai/technologies/>

Other topics

доклад о разметчиках https://youtu.be/gHi6IFUs6_Q

Thanks for attention!

Questions?

